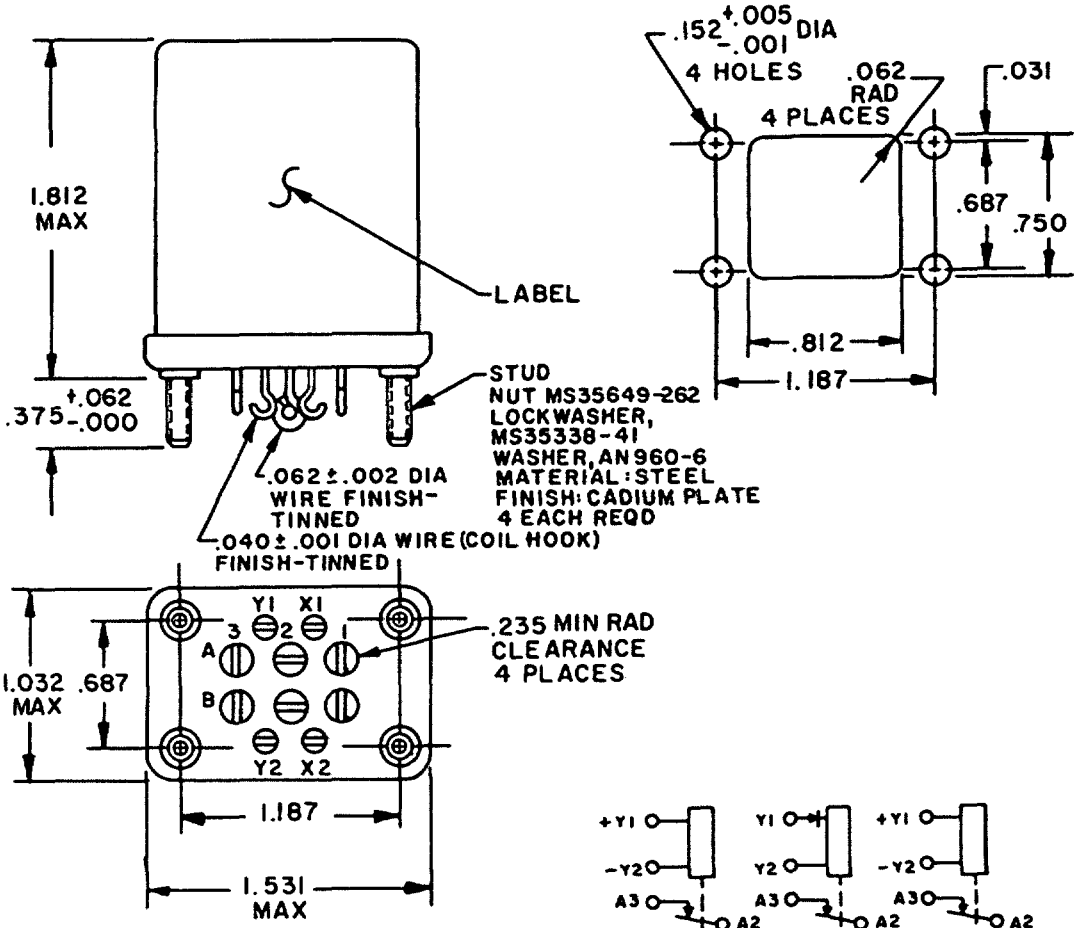
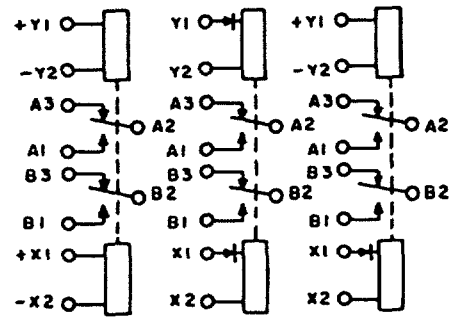
FED. SUP CLASS
5945

INACTIVE FOR NEW DESIGN AFTER 5 JUN 87
NO SUPERSEDING STANDARD
(FOR NEW DESIGN USE MIL-R-6106/50)



Inches	mm
.000	0.00
.001	0.03
.002	0.05
.005	0.13
.031	0.79
.040	1.02
.062	1.57
.152	3.86
.187	4.75
.235	5.97
.375	9.53
.687	17.45
.750	19.90
.812	20.62
1.032	26.21
1.531	38.89
1.812	46.02



Ⓓ ENTIRE STANDARD REVISED

APPROVED 1 Mar 1963 REVISED Ⓓ 5 JUN 87

User activities: Army - Navy - Air Force -
Review activities: Army - EC Navy - Air Force - 11

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

P.A	International Interest	TITLE	MILITARY STANDARD
AIR FORCE - 85 Other Cust Navy - AS		RELAYS, ELECTROMAGNETIC, 10 AMPERES, 2 PDT, TYPE I, MAGNETIC LATCH, STUD MOUNTED, SOLDER HOOKS, HERMETICALLY SEALED	MS25466
Procurement Specification		SUPERSEDES:	PAGE 1 OF 5

945-0745-20

DD FORM 672 1 MAY 73 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.
AMSC N/A

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010.
4. Terminal numbers need not appear on relay header provided there is affixed to the relay a suitable legible circuit diagram that permanently and positively identifies each terminal location specified hereon.
5. The use of diodes on ac relays is optional. Actual application must be shown on label.
6. Shock, vibration, and acceleration requirements applicable with coils de-energized.
7. Relay is magnetically latched in both positions. Caution note to observe polarity must appear on relays with dc coils.
8. Caution note to observe polarity must appear on relays with dc coils.
9. In the event of a conflict between the test of this standard and the references cited herein, the text of this standard shall take precedence.
10. Referenced government documents of the issue listed in that issue of the department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.

TABLE I. Dash numbers and characteristics.

Dash number MS 25466-	Type	Coil	Terminal type	Mounting or mating socket	Max weight in pounds
D1	I	dc	Solder hook	Stud	0.30
A1	I	ac	Solder hook	Stud	0.32
AD1	I	ac-dc	Solder hook	Stud	0.32

User activities: Army - Navy - Air Force -
Review activities: Army - Navy - EC Air Force - 11
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APPROVED 1 Mar 1963 REVISED (D) ENTIRE STANDARD REVISED

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Review activities: Army - Navy - EC Air Force -11

User activities: Army - Navy - Air Force -

P.A AIR FORCE-85 Other Cust NAVY-AS

International interest

Procurement Specification MIL-R-6106

SUPERSEDES:

TITLE RELAYS, ELECTROMAGNETIC, 10 AMPERES, 2 PDT, TYPE I, MAGNETIC LATCH, STUD MOUNTED, SOLDER HOOKS, HERMETICALLY SEALED

MILITARY STANDARD

MS25466

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FED. SUP CLASS 5945

TABLE II. Operating characteristics.

MS part no. MS25466-	Coil data											Time - (milliseconds maximum)				
	Coil	Nominal			Max		Max pick-up voltage			Drop-out voltage	Operate 3/	Release 4/	Contact bounce			
													Main		Aux	
		Volts 1/	Freq. Hz	Res Ω	Volts	Amperes	Normal 3/	High temp test	Cont current test				NO	NC	NO	NC
D1	X1, X2 Y1, Y2	28	dc	N/A	29	0.17	18	18	19.8	N/A	25	N/A	2	2		
A1	X1, X2 Y1, Y2	115	400 5/	N/A	122	0.07	90	90	95	N/A	25	N/A	2	2		
AD1	X1, X2	115	400 2/	N/A	122	0.07	90	90	95	N/A	25	N/A	2	2		
	Y1, Y2	28	dc	N/A	29	0.17	18	18	19.8	N/A	25	N/A	2	2		

1/ CAUTION: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ MS25466-A1 and -AD1 may be used on 60 Hz if maximum ambient temperature is limited to +85°C (maximum coil current shall be 0.077 ampere).
3/ Over the temperature range.
4/ With nominal coil voltage.
5/ From nominal coil voltage.

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Review activities: Army - Navy - EC Air Force - 11

User activities: Army - Navy - Air Force -

TABLE III. Rated contact load (amperes per pole) (case grounded).

Type of load	Life operating cycles X 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/				See appropriate notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive	100	10	10			10	6			10	6			
Inductive	100													
Inductive	20	6	6			10	4			10	4			
Motor	100	4	4			4	3			4	3			
Lamp	100	2	2			2	1.5			2	1.5			
Transfer load														2/
Mechanical life reduced current	400	2.5	2.5			2.5	2			2.5	2			
Intmd current	Applicable per specification													

1/ Absence of value indicates relay is not rated for 3-phase applications.
2/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.

P.A AIR FORCE-85 Other Cust NAVY-AS

International interest

Procurement Specification MIL-R-6106

SUPERSEDES:

TITLE RELAYS, ELECTROMAGNETIC, 10 AMPERES, 2 PDT, TYPE I, MAGNETIC LATCH, STUD MOUNTED, SOLDER HOOKS, HERMETICALLY SEALED

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MILITARY STANDARD

MS25466

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FED. SUP CLASS 5945

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Review activities: Army - Navy - EC Air Force - 11

User activities: Army - Navy - Air Force -

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Procurement Specification MIL-R-6106

P.A AIR FORCE-85 Other Cust Navy - AS

International interest

SUPERSEDES:

TITLE RELAYS, ELECTROMAGNETIC, 10 AMPERES, 2 PDT, TYPE I, MAGNETIC LATCH, STUD MOUNTED, SOLDER HOOKS, HERMETICALLY SEALED

MILITARY STANDARD MS25466

PAGE 5 OF 5

FED. SUP CLASS 5945

Environmental characteristics

Temperature range	-70°C to +125°C
Max altitude rating	80,000 ft
Shock G-level	50 G
Duration	11 ms
Max duration contact opening	10 µs
Vibration - sinusoidal	
G-level	10 G
Frequency range	5 - 1500 Hz
Acceleration	15 G

Electrical characteristics

Insulation resistance, initial	100 megohms
After life or environmental tests	50 megohms

Dielectric strength (sea level)

	Initial	After life tests
Coil to case	1,000 V rms	1,000 V rms
Aux contacts		
All other points	1,500 V rms	1,125 V rms

Dielectric strength (altitude) (When mounted in mating socket) 80,000 ft

Coil to case		250 V rms
Aux contacts		
All other points		350 V rms

Max contact drop initial	0.150 volt
After life test	0.175 volt
Overload current	40 amperes dc, 60 amperes ac
Rupture current	50 amperes dc, 80 amperes ac
Duty rating	Continuous
RFI specification	MIL-STD-461

(Applicable to coil circuits of ac operated relays)

Quality conformance inspection

Performance of groups B and C testing are not applicable.

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